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Catheter position indicator and related systems and method

Abstract

A catheter assembly may include a catheter adapter having a catheter extending from its distal end. The catheter may include a proximal end, a tip, and a length of tubing therebetween. The length of tubing may include one or more position indicators to indicate a position of the catheter relative to a vasculature. The catheter assembly may include an introducer needle extending through the length of tubing and comprising a sharp distal tip configured to introduce the catheter into the vasculature.

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Background/Summary

RELATED APPLICATIONS (1) This application claims benefit of U.S. Provisional Patent Application No. 63/055,210, filed on Jul. 22, 2020, entitled CATHETER POSITION INDICATOR AND RELATED SYSTEMS AND METHODS, which is incorporated herein in its entirety.

BACKGROUND
(1) Catheters are commonly used for a variety of infusion therapies. For example, catheters may be used for infusing fluids, such as normal saline solution, various medicaments, and total parenteral nutrition, into a patient. Catheters may also be used for withdrawing blood from the patient.
(2) A common type of catheter is an over-the-needle peripheral intravenous catheter. As its name implies, the over-the-needle catheter may be mounted over an introducer needle having a sharp distal tip. The catheter and the introducer needle may be assembled so that the distal tip of the introducer needle extends beyond the distal tip of the catheter with the bevel of the needle facing away from skin of the patient. The catheter and the introducer needle are generally inserted at a shallow angle through the skin into vein of the patient.
(3) In order to verify proper placement of the introducer needle and/or the catheter in the blood

vessel, a clinician generally confirms that there is blood "flashback." Thus, the clinician may visualize the blood and thereby confirm placement of the introducer needle within the vasculature. Once placement of the needle has been confirmed, the clinician may temporarily occlude flow in the vein and remove the introducer needle, leaving the catheter in place and secured for future blood withdrawal and/or fluid infusion.

(4) In some cases, however, the catheter may become displaced or "dislodged" due to improper securement or, more commonly, due to forces that are greater than the securement method was designed to withstand. For example, catheter dislodgement may occur when a patient rolls over in bed, catches their line on a bed rail, transfers between beds, fidgets, or intentionally pulls out their line.

(5) At minimum, catheter dislodgement may necessitate a restart of the intravenous catheter, which places clinicians at an increased risk due to potential sharps injuries as well as blood or drug exposure. In addition, an intravenous catheter restart may be both inconvenient and uncomfortable for the patient. In more serious cases, such as where dislodgement causes loss of peripheral intravenous site integrity, the patient may need to be fit with a more invasive central line with increased risks of phlebitis, infiltration, and infection, as well as longer treatment times. Potential worst-case scenarios may include bleed outs, air embolisms, and bloodstream infections and associated risks of patient death.

(6) The subject matter claimed herein is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example technology area where some implementations described herein may be practiced.

SUMMARY

(7) The present disclosure relates generally to vascular access devices and related systems and methods. As previously discussed, dislodgement may occur due to improper securement or, more commonly, dislodgement caused from patients rolling over in bed, catching their lines on bed rails, transferring to a different bed, fidgeting, or intentionally pulling on their lines, for example.

(8) Such dislodgement may necessitate a line restart, which tends to be both inconvenient and uncomfortable. In some cases, the patient may need to be fit with a more invasive central line, associated with increased risks of phlebitis, infiltration, infection, as well as longer treatment times. Worst-case scenarios may include bleed outs, air embolisms, and bloodstream infections with increased risk of patient death. Embodiments described herein may provide an early indication of catheter movement to thereby reduce a risk of dislodgement and its associated complications.

(9) In some embodiments, a catheter assembly may include a catheter adapter having a proximal end, a distal end, and a lumen extending therebetween. In some embodiments, a catheter may extend from the distal end. In some embodiments, the catheter may include a proximal end, a tip, and a length of tubing therebetween. In some embodiments, the length of tubing may include at least one position indicator to indicate a position of the catheter relative to a vasculature of a patient.

(10) Some embodiments may include an introducer needle extending through the length of tubing. In some embodiments, the introducer needle may include a sharp distal tip configured to introduce the catheter into the vasculature.

(11) In some embodiments, the at least one position indicator may include at least one marking disposed on an outer surface of the length of tubing. Some embodiments of the at least one marking may include a colored band, a tick mark, or a graduated mark. In some embodiments, the at least one marking comprises a plurality of non-uniformly spaced markings. In other embodiments, the at least one marking comprises a plurality of uniformly spaced markings.

(12) In some embodiments, a distance between non-uniformly spaced markings may decrease proximate to the distal end of the catheter adapter. In some embodiments, the at least one marking may indicate a distance to the distal end of the catheter. In some embodiments, the at least one

marking may indicate dislodgement of the catheter relative to the vasculature. Some embodiments of the at least one position indicator may include a first position indicator located near the tip of the catheter and a second position indicator located near the proximal end of the catheter.

(13) In some embodiments, a vascular access device to detect a catheter position within a vasculature may include a catheter comprising a proximal end, a tip, and a length of tubing therebetween. In some embodiments, a position indicator may be disposed on an outer surface of the length of tubing and may be configured to indicate a position of the catheter within a vasculature.

(14) In some embodiments, the position indicator may include at least one marking printed onto the outer surface of the length of tubing. In some embodiments, the position indicator may include a plurality of markings. In some embodiments, each of the plurality of markings is uniformly spaced.

(15) In some embodiments, each of the plurality of markings may include a non-uniform characteristic. The non-uniform characteristic may include, for example, color, width, length, and/or spacing. In some embodiments, the position indicator may be a colored band, a tick mark, or a graduated mark.

(16) In some embodiments, the position indicator may include a first plurality of markings located near the tip of the catheter and a second plurality of markings located near the proximal end of the catheter. In some embodiments, each of the first plurality of markings may be uniformly spaced from each other of the first plurality of markings. Likewise, each of the second plurality of markings may be is uniformly spaced from each other of the second plurality of markings.

(17) Some embodiments may include a method for detecting catheter position. In some embodiments, the method may include providing a catheter assembly to introduce a catheter into a vasculature of a patient. Some embodiments of the catheter assembly may include a catheter adapter comprising a proximal end, a distal end, and a lumen extending therebetween. The catheter may extend from the distal end. In some embodiments, the catheter may include a proximal end, a tip, and a length of tubing therebetween. Some embodiments of the length of tubing may include a position indicator to indicate a position of the catheter relative to the vasculature.

(18) In some embodiments, an introducer needle may extend through the length of tubing. Some embodiments of the introducer needle may include a sharp distal tip to introduce the catheter into the vasculature at an intravenous entry site of the patient. In some embodiments, the catheter may be secured such that the position indicator is aligned with the intravenous entry site. Some embodiments of the method may further include automatically monitoring a location of the position indicator relative to the intravenous entry site to detect catheter dislodgement.

(19) In some embodiments, the position indicator may include a plurality of markings printed onto an outer surface of the length of tubing. In some embodiments, the method may further include automatically activating an alert in response to the location of the position indicator misaligning with the intravenous entry site.

(20) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the present disclosure, as claimed. It should be understood that the various embodiments are not limited to the arrangements and instrumentality shown in the drawings. It should also be understood that the embodiments may be combined, or that other embodiments may be utilized and that structural changes, unless so claimed, may be made without departing from the scope of the various embodiments of the present disclosure. The following detailed description is, therefore, not to be taken in a limiting sense.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) Example embodiments will be described and explained with additional specificity and detail

through the use of the accompanying drawings in which:
(2) FIG. **1**A is an upper perspective view of an example catheter assembly, according to some embodiments;
(3) FIG. **1**B is a cross-sectional view of the catheter assembly of FIG. **1**A, according to some embodiments;
(4) FIG. **2** is an upper perspective view of an example catheter illustrating multiple position indicators according to some embodiments;
(5) FIG. **3**A is an upper perspective view of an example catheter, according to some embodiments;
(6) FIG. **3**B is an upper perspective view of an example catheter, according to some embodiments;
(7) FIG. **3**C is an upper perspective view of an example catheter, according to some embodiments;
(8) FIG. **3**D is an upper perspective view of an example catheter, according to some embodiments;
(9) FIG. **4** is an upper perspective view of an example catheter of the catheter assembly of FIG. **1**A, according to some embodiments; and
(10) FIG. **5** is an upper perspective view of the catheter assembly of FIG. **1**A disposed in a vasculature according to some embodiments.

DESCRIPTION OF EMBODIMENTS

(11) As used in this specification, the term "distal" refers to a direction away from a clinician who would place the device into contact with a patient, and nearer to the patient. The term "proximal" refers to a direction nearer to the clinician who would place the device into contact with the patient, and farther away from the patient. Thus, for example, the end of a catheter first touching the body of the patient is the distal end, while the opposite end of the catheter is the proximal end of the catheter.
(12) As previously mentioned, the catheter may become dislodged due to improper securement or an intentional or unintentional application of force to the catheter or catheter adapter. For example, catheter dislodgement may occur when a patient rolls over in bed, catches their line on a bed rail, transfers between beds, fidgets, or intentionally pulls out their line.
(13) Catheter dislodgement may necessitate a line restart, placing a clinician at an increased risk due to potential sharps injury as well as exposure to blood and/or drugs. An intravenous catheter restart may also be both inconvenient and uncomfortable for the patient. In more serious cases, such as where dislodgement causes loss of peripheral intravenous site integrity, the patient may need to be fit with a more invasive central line with increased risks of phlebitis, infiltration, and infection, as well as longer treatment times. Potential worst-case scenarios may include bleed outs, air embolisms, and bloodstream infections and associated risks of patient death. Embodiments described herein may provide an early indication of catheter displacement or dislodgement to enable a clinician to immediately address the issue without requiring a line restart.
(14) Referring now to FIGS. **1**A-**1**B, in some embodiments, a catheter system **10** may include a needle assembly **12** and a catheter assembly **14**. FIGS. **1**A-**1**B illustrate the catheter system in an insertion position, ready for insertion into a vein of a patient, according to some embodiments. In some embodiments, the catheter assembly **14** may include a catheter adapter **16**, which may include a distal end **18**, a proximal end **20**, and a lumen **21** extending therebetween. In some embodiments, the catheter **22** may extend from the distal end **18**. Some embodiments of a catheter **22** may include a peripheral intravenous catheter ("PIVC"). In some embodiments, the catheter **22** may include a proximal end **26**, a tip **24**, and a length of tubing therebetween. In some embodiments, the proximal end **26** of the catheter **22** may be secured within the catheter adapter **16**.
(15) In some embodiments, the needle assembly **12** may include a needle hub **28**, which may be removably coupled to the catheter adapter **16**. In some embodiments, the needle assembly **12** may include an introducer needle **30**. In some embodiments, a proximal end of the introducer needle **30** may be secured within the needle hub **28**. In some embodiments, the introducer needle **30** may extend through the length of tubing when the catheter **22** is ready for insertion into the vein of a patient, as illustrated, for example, in FIGS. **1**A-**1**B.

(16) In some embodiments, the needle assembly **12** may include a needle grip **32**, which a clinician may grip and move proximally to withdraw the introducer needle **30** from the vein once placement of the catheter **22** within the vein is confirmed. In some embodiments, the catheter assembly **14** may include an extension tube **34**. In some embodiments, a distal end of the extension tube **34** may be coupled to the catheter adapter **16** and a proximal end of the extension tube **34** may be coupled to an adapter **36**.
(17) In some embodiments, a fluid infusion device may be coupled to the adapter **36** to deliver fluid to the patient via the catheter **22** inserted in the vein when the introducer needle **30** is removed from the catheter assembly **14**. In some embodiments, a blood collection device may be coupled to the adapter **36** to withdraw blood from the patient via the catheter **22** inserted in the vein.
(18) In some embodiments, the catheter assembly **14** may be an integrated catheter system **10**, having the extension tube **34** integrated within the catheter adapter **16** such as, for example, the BD NEXIVA™ Closed IV Catheter System, the BD NEXIVA™ DIFFUSICS™ Closed IV Catheter System, the BD PEGASUS™ Safety Closed IV Catheter System, or another integrated catheter system. An example of an integrated catheter system **10** is illustrated in FIGS. **1**A-**1**B. In some embodiments, the catheter assembly **14** may be non-integrated, without the extension tube **34**.
(19) In some embodiments, the catheter assembly **14** may be vented to observe blood and facilitate proximal flow of blood within the introducer needle **30** and/or the catheter **22**. In some embodiments, the catheter assembly **14** may be vented in any suitable manner. For example, a vent plug **38** may be coupled to the adapter **36** during insertion of the catheter assembly **14** into the patient. In some embodiments, the vent plug **38** may be permeable to air but not to blood. In some embodiments, the catheter **22**, the catheter adapter **16**, the extension tube **34**, the adapter **36**, and the vent plug **38** may be in fluid communication.
(20) In some embodiments, blood may flow through a sharp distal tip **40** of the introducer needle **30** in response to the introducer needle **30** being inserted into a vein of the patient. The blood flow may cause “flashback,” or give an indication to the clinician that the introducer needle **30** has entered the vein. After confirming the sharp distal tip **40** is positioned within the vein of the patient, the clinician may lower an insertion angle and advance the catheter **22** a short distance to ensure the distal end **24** of the catheter **22** is in the vein before threading the catheter **22** off the introducer needle **30**. The distal end **24** of the catheter **22** may be separated from a proximal end of the sharp distal tip **40** of the introducer needle **30** by a distance called “lie distance.” In some embodiments, a length of a bevel extending from the sharp distal tip **40** may vary by gauge size of the introducer needle **30**. Accordingly, in some embodiments, a distance the catheter **22** needs to travel from the sharp distal tip **40** to enter the vein may vary by gauge and lie distance. Catheter **22** insertion may fail in the event the catheter **22** is threaded off the introducer needle **30** too soon.
(21) In some embodiments, the length of tubing of the catheter **22** may include a position indicator **42** to indicate a position of the catheter **22** relative to a vasculature of a patient. The position indicator **42** may be one or more markings **44** disposed on an outer surface of the length of tubing. Some embodiments of the one or more markings **44** may include one or more colored bands, tick marks, or graduated marks. In some embodiments, the marking **44** may include a plurality of non-uniformly spaced markings **44**. In other embodiments, the marking **44** may include multiple uniformly spaced markings **44**.
(22) In some embodiments, the position indicator **42** may include a plurality of markings **44**, which may indicate to the clinician a depth of insertion of the catheter **22** into the vein. In some embodiments, the spacing between the markings **44** may be a fraction of the lie distance. In some embodiments, the lie distance may be gauge dependent. In some embodiments, the clinician may advance the catheter **22** several markings **44** before threading off the catheter **22** from the introducer needle **30**. In some embodiments, a distance between non-uniformly spaced markings **44** may decrease proximate to the distal end **18** of the catheter adapter **16**. In some embodiments, the one or more markings **44** may indicate a distance to the distal end **24** of the catheter **22**. In some

embodiments, the markings **44** may indicate dislodgement of the catheter **22** relative to the vasculature. Some embodiments of the position indicator **42** may include a first position indicator **42** located near the tip **24** of the catheter **22** and a second position indicator located near the proximal end **26** of the catheter **22**. In some embodiments, the first position indicator **42** may include a first plurality of markings **44** uniformly spaced from each other. Likewise, each of the second position indicator **42** may include a second plurality of markings **44** uniformly spaced with respect to each other.

(23) Referring now to FIG. **2**, in some embodiments, the position indicator **42** may include markings **44** printed on the surface of the length of tubing of the catheter **22**. In some embodiments, the position indicator **42** may include patterns or material applied to the surface or bonded to the surface. In some embodiments, the markings **44** may be pad printed or etched onto the length of tubing. In other embodiments, the markings **44** may be applied by any other suitable method such as screen printing, digital printing, flexography printing, UV-Litho printing, or laser printing. In some embodiments, the markings **44** may be printed in several colors or patterns.

(24) In some embodiments, each of the markings **44** may include a non-uniform characteristic. The non-uniform characteristic may include, for example, color, width, length, and/or spacing. In some embodiments, the markings **44** may be staggered or spaced apart in the longitudinal axial direction. In these embodiments, the placement of the markings **44** may facilitate identification by the clinician of a depth at which the tip **24** of the catheter **22** is inserted. In these embodiments, the markings **44** may facilitate confirmation of one or more positions of the catheter tip **24** with respect to the vasculature or surface of skin.

(25) For example, in some embodiments, the markings **44** may indicate that the catheter **22** has entered the vein, that the catheter **22** has been further fed into the vein, and/or that the catheter **22** has been fully inserted into the vein. In some embodiments, the markings **44** may be configured to indicate the tip **24** of the catheter **22** is within the vein. In some embodiments, the markings **44** may be configured to indicate the tip **24** of the catheter **22** is withdrawn or partially withdrawn from the vein. In some embodiments, the markings **44** may facilitate visualization of blood flashback by the clinician, thereby confirming insertion of the introducer needle **30** within the vasculature.

(26) Referring now to FIG. **3**A, in some embodiments, the markings **44** may be colored bands **46**. In some embodiments the colored bands **46** may be of uniform width. In other embodiments, the colored bands **46** may be a non-uniform width. Because it is often difficult to determine a position or depth of the catheter **22** at a glance, contrasting colors may be a particularly useful. As the catheter **22** is inserted further into the vasculature or vein, the color that is visible at the surface of the skin may provide a more intuitive indication or reference of the depth of insertion of the catheter **22** into the vein. In some embodiments, the visible color may indicate dislodgement of the catheter **22** after it has been previously positioned and secured, which will be explained in more detail below. In some embodiments, the colors and width of the bands of the position indicator **42** may vary depending on the application of the catheter **22**.

(27) Referring now to FIG. **3**B, in some embodiments, the plurality of markings **44** may be uniform or evenly spaced apart. In some embodiments, the markings **44** may include lines or tick marks **44** generally perpendicular to a longitudinal axis of the catheter **22**. In some embodiments, the tick marks **44** may be uniform or may alternate between major and minor tick marks **44** to more accurately indicate the depth of the catheter **22** and help to provide a clear visual reference. In some embodiments, the thickness and spacing of the marks **44** may vary according to the application of the position indicator. In some embodiments, the clinician may insert the catheter **22** such that a particular marking **44** is aligned or even with an outer surface of skin of the patient. In some embodiments, the particular marking **44** or color band **46** may be aligned with skin of the patient to indicate to the clinician that the catheter **22** has been properly inserted into the vasculature or vein.

(28) Referring now to FIGS. **3**C and **3**D, in some embodiments, the position indicator **42** may include more than one plurality of markings **44**. In some embodiments, one or more pluralities of

markings **44** may be located near the catheter tip **24**. Likewise, in some embodiments, one or more pluralities of markings **44** may be located near the proximal end **26** of the catheter **22**. In these and other embodiments, the pluralities of markings **44** may be uniformly or non-uniformly spaced.
(29) Referring now to FIG. **4**, in some embodiments, the position indicator **42** may include a first plurality of markings **48** and a second plurality of markings **50**. In some embodiments, the first plurality of markings **48** may be located near the catheter tip **24** and the second plurality of markings may be located near the proximal end **26** of the catheter **22**. In some embodiments, the first plurality of markings **48** and the second plurality of markings **50** may be uniformly spaced. In other embodiments, the first plurality of markings **48** and the second plurality of markings **50** may be non-uniformly spaced. In at least one embodiment, the first plurality of markings **48** may include two marks **44** spaced apart by a first distance **52**. In at least one embodiment, the first distance **52** may be approximately 2-3 mm. The second plurality of markings **50** may also include two marks **44** spaced apart by a second distance **54**. In some embodiments, the second distance **54** may also be approximately 2-3 mm. In some embodiments, the first plurality of markings and the second plurality of markings may be spaced apart by a third distance **56**. In one embodiment, the third distance **56** may be approximately 5-10 mm.
(30) In some embodiments, the first plurality of markings **48** may provide a reference for depth of insertion of the catheter **22** and the second plurality of markings **50** may provide a reference for threading the catheter **22** into the vein. In some embodiments, the second plurality of markings **50** may provide a reference for determining whether the catheter **22** has become dislodged. In some embodiments, the first plurality of markings **48** may be spaced apart from each other such that the first distance **52** is half the length of the needle bevel. In some embodiments, the first distance **52** may be the catheter lie distance. In some embodiments, the first distance **52** may be the length of the catheter taper **58**. In some embodiments, the second plurality of markings **50** may be located at a distance about 1/10 of the length of the catheter **22** measured from the catheter proximal end **26**.
(31) Referring now to FIG. **5**, in some embodiments, the position indicator **42** may indicate catheter dislodgement. In some embodiments, catheter dislodgement may be indicated by an exposed marking **44** that was not previously visible. In other embodiments, catheter dislodgement may be indicated by variation in the location of the marking **44** relative to the surface of skin. For example, after the clinician advances the catheter **22** several markings **44** within the vein prior to threading off the catheter **22** from the introducer needle **30**, the marking **44** visible at the surface of the skin may be noted in the medical record, on the catheter assembly **14**, or in other suitable location. In some embodiments, the notation may indicate the marking **44** by color band visible at the surface of or under the skin, a particular tick mark, a reference location, or a count of the number of markings **44** exposed.
(32) In one example, a catheter **22** may include two markings **44** located near the proximal end **26** of the catheter **21**. In some embodiments, the catheter **22** may be inserted into the vein and secured such that one of the markings **44** is exposed while another marking **44** is under the surface of the skin. In some embodiments, if both markings **44** are exposed, the clinician may be alerted to possible dislodgement and may conduct further investigation to determine an appropriate response.
(33) Some embodiments may include a method for detecting catheter position. In some embodiments, the method may include providing a catheter assembly to introduce a catheter into a vasculature of a patient. Some embodiments of the catheter assembly may include a catheter adapter comprising a proximal end, a distal end, and a lumen extending therebetween. The catheter may extend from the distal end. In some embodiments, the catheter may include a proximal end, a tip, and a length of tubing therebetween. Some embodiments of the length of tubing may include a position indicator to indicate a position of the catheter relative to the vasculature.
(34) In some embodiments, an introducer needle may extend through the length of tubing. Some embodiments of the introducer needle may include a sharp distal tip to introduce the catheter into the vasculature at an intravenous entry site of the patient. In some embodiments, the catheter may

be secured such that the position indicator is aligned with the intravenous entry site. Some embodiments of the method may further include automatically monitoring a location of the position indicator relative to the intravenous entry site to detect catheter dislodgement.

(35) In some embodiments, the position indicator may include a plurality of markings printed onto an outer surface of the length of tubing. In some embodiments, the method may further include automatically activating an alert in response to the location of the position indicator misaligning with the intravenous entry site.

(36) Some embodiments may include a method to detect catheter **22** position. The method may include providing the catheter assembly **14** to introduce the catheter **22** into the vasculature. In some embodiments, the catheter assembly **14** may include a catheter adapter **16** comprising the proximal end **20**, the distal end **18**, and the lumen **21** extending therebetween. In some embodiments, the catheter **22** may extend from the distal end **18**. The catheter **22** may include the proximal end **26**, the tip **24**, and the length of tubing therebetween. In some embodiments, the length of tubing may include the position indicator **42**, which may indicate a depth of insertion of the catheter **22** into the vasculature or vein.

(37) In some embodiments, the position indicator may include a marking **44** disposed on an outer surface of the length of tubing of the catheter **22**. In some embodiments, an introducer needle **30** may extend through the length of tubing. Some embodiments of the introducer needle **30** may include a sharp distal tip **40** to introduce the catheter **22** into the vasculature at an intravenous entry site of the patient. In some embodiments, the catheter **22** may be advanced into the vasculature and secured such that the position indicator **42** is aligned with the intravenous entry site. In some embodiments, the catheter **22** may be advanced into the vasculature by several markings **44** before threading off the catheter **22** from the introducer needle **30**.

(38) In some embodiments, the catheter **22** may be inserted such that a first plurality of markings **48** is positioned within the vasculature and a second plurality of markings **50** is aligned or even with an outer surface of skin of the patient. In some embodiments, in response to a second of the markings **50** being aligned with the outer surface of the skin of the patient, the catheter **22** may be inserted to a second depth which may correspond to full insertion of the catheter **22** within the vein. In some embodiments, the second plurality of markings **50** may be proximal to the surface of skin of the patient. In some embodiments, the position indicator **42** may include a plurality of markings **44** spaced approximately 2-3 mm apart located at a distance about 1/10 of the catheter **22** length from the proximal end.

(39) In some embodiments, the catheter adapter **14** may be secured to the patient. In some embodiments, variation in a position of one or more markings **44** relative to the surface of skin may indicate a potential catheter dislodgement. In this manner, some embodiments may facilitate early detection of catheter **22** dislodgment and enable corrective action to be taken.

(40) Some embodiments of the method may further include automatically monitoring a location of the position indicator **42** relative to the intravenous entry site to detect catheter **22** dislodgement. In some embodiments, an image processor **62** or other device **60** may automatically monitor the location of the position indicator **42** relative to the catheter insertion site. In some embodiments, the image processor may be coupled to the proximal end **20** of the catheter adapter **16** and/or secured to the patient.

(41) In some embodiments, the image processor **62** may be configured to record one or more images of the insertion site relative to the markings **44**. In some embodiments, the image processor may also be configured to analyze the relative locations of the catheter **22** and/or the markings **44** to detect dislodgement of the catheter **22** and automatically activate an alert.

(42) In some embodiments, the image processor **62** may detect the marking **44** and capture the image by any suitable means. In some embodiments, the image processor **62** may be configured to detect color contrast of the exposed markings **44**. Embodiments of the image processor **62** may be combined with any other embodiments disclosed herein to facilitate placement of the catheter **22**

within the vasculature and/or detection of catheter **22** dislodgement.

(43) In some embodiments, in response to a determination that the catheter **22** has been dislodged, an alert **64** may be activated. In some embodiments, the alert may be located on the image processor **62** or any other suitable device **60**. In some embodiments, in response to the tip **24** of the catheter **22** being dislodged or fully or partially withdrawn, or in response to the position indicator **42** misaligning with the intravenous entry site, the alert **64** may be automatically activated. In some embodiments, the alert **64** may be activated locally or remotely. In some embodiments, the alert **64** may include a visual, haptic, and/or audible alert. In some embodiments, one or more alerts **64** may change or stop in response the introducer needle **30** being removed from the vein, the catheter **22** or catheter tip **24** being removed from the vein, or the catheter **22** being at least partially withdrawn from the vein.

(44) All examples and conditional language recited herein are intended for pedagogical objects to aid the reader in understanding the present disclosure and the concepts contributed by the inventor to furthering the art and are to be construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present disclosure have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the present disclosure.

Claims

1. An over-the-needle peripheral intravenous catheter assembly, comprising: a catheter adapter comprising a proximal end, a distal end, and a lumen extending therebetween; an over-the-needle peripheral intravenous catheter extending from the distal end of the catheter adapter, the over-the-needle peripheral intravenous catheter comprising a proximal end, a tip, and a length of tubing therebetween, wherein the length of tubing comprises at least one position indicator to indicate a position of the over-the-needle peripheral intravenous catheter relative to a vasculature, wherein the at least one position indicator comprises a first pair of markings and a second pair of markings, wherein the first pair of markings is disposed on an outer surface of the length of tubing closer to the tip of the over-the-needle peripheral intravenous catheter than the distal end of the catheter adapter, wherein the second pair of markings is disposed on the outer surface of the length of tubing closer to the distal end of the catheter adapter than the tip of the over-the-needle peripheral intravenous catheter, wherein a first marking and a second marking of the first pair of markings are spaced apart, wherein a first marking and a second marking of the second pair of markings are spaced apart, wherein the first pair of markings provides a reference for depth of insertion of the over-the-needle peripheral intravenous catheter and the second pair of markings provides an indicator of catheter dislodgement by variation in a location of the first marking or the second marking of the second pair of markings with respect to skin of a patient, wherein an unmarked portion of the tubing extends from the first pair of markings to the second pair of markings; and an introducer needle extending through the length of tubing of the over-the-needle peripheral intravenous catheter and comprising a sharp distal tip configured to introduce the over-the-needle peripheral intravenous catheter into the vasculature, wherein a distance between the first marking and the second marking of the first pair of markings is a lie distance, wherein the lie distance equals a distance between a proximal end of a bevel of the introducer needle and the tip of the over-the-needle peripheral intravenous catheter.
2. The over-the-needle peripheral intravenous catheter assembly of claim 1, wherein the first pair of markings or the second pair of markings is selected from the group consisting of colored bands, tick marks, and graduated marks.
3. The over-the-needle peripheral intravenous catheter assembly of claim 1, wherein the first pair of markings and the second pair of markings are uniformly spaced.
4. The over-the-needle peripheral intravenous catheter assembly of claim 1, wherein the second

pair of markings is located at a distance from the proximal end of the catheter of about 1/10 of a length of the catheter.
5. The over-the-needle peripheral intravenous catheter assembly of claim 1, wherein a distance between the first marking and the second marking of the first pair of markings is half of a length of a bevel of the introducer needle.
6. The over-the-needle peripheral intravenous catheter assembly of claim 1, wherein the first pair of markings includes exactly two markings.
7. The over-the-needle peripheral intravenous catheter assembly of claim 1, wherein the first marking and the second marking of the first pair of markings are spaced apart by a length of a taper at the tip of the over-the-needle peripheral intravenous catheter.
8. The over-the-needle peripheral intravenous catheter assembly of claim 1, further comprising an image processor coupled to the over-the-needle peripheral intravenous catheter assembly, wherein the image processor is configured to record an image of an insertion site of the over-the-needle peripheral intravenous catheter assembly and analyze a location of the second pair of markings to detect dislodgement of the over-the-needle peripheral intravenous catheter.
9. The over-the-needle peripheral intravenous catheter assembly of claim 8, further comprising an alert, wherein in response to the image processor detecting the dislodgement, the alert is configured to be activated.
10. An over-the-needle peripheral intravenous catheter assembly, comprising: an over-the-needle peripheral intravenous catheter comprising a proximal end, a tip, and a length of tubing therebetween; and an introducer needle extending through the over-the-needle peripheral intravenous catheter, wherein the introducer needle comprises a bevel; a position indicator disposed on an outer surface of the length of tubing, the position indicator configured to indicate a position of the over-the-needle peripheral intravenous catheter within a vasculature, wherein the position indicator comprises a first pair of markings disposed closer to the tip of the over-the-needle peripheral intravenous catheter than the proximal end of the over-the-needle peripheral intravenous catheter, wherein the first pair of markings provides a reference for depth of insertion of the over-the-needle peripheral intravenous catheter into the vasculature, wherein a first marking and a second marking of the first pair of markings are spaced apart by a lie distance, wherein the lie distance is equal to a distance between a proximal end of the bevel and the tip of the over-the-needle peripheral intravenous catheter.
11. The over-the-needle peripheral intravenous catheter assembly of claim 10, wherein the position indicator is printed onto the outer surface of the length of tubing.
12. The over-the-needle peripheral intravenous catheter assembly of claim 10, wherein the first pair of markings is spaced apart by 2-3 mm.
13. The over-the-needle peripheral intravenous catheter assembly of claim 12, wherein the first pair of markings comprises a non-uniform characteristic selected from the group consisting of color, width, and length.
14. The over-the-needle peripheral intravenous catheter assembly of claim 10, wherein each of the first pair of markings is selected from the group consisting of colored bands, tick marks, and graduated marks.